CHANDIGARH UNIVERSITY

APEX INSTITUTE OF TECHNOLOGY

ASSIGNMENT 1

Subject: Computer Networks
Subject Code: 24CST-339

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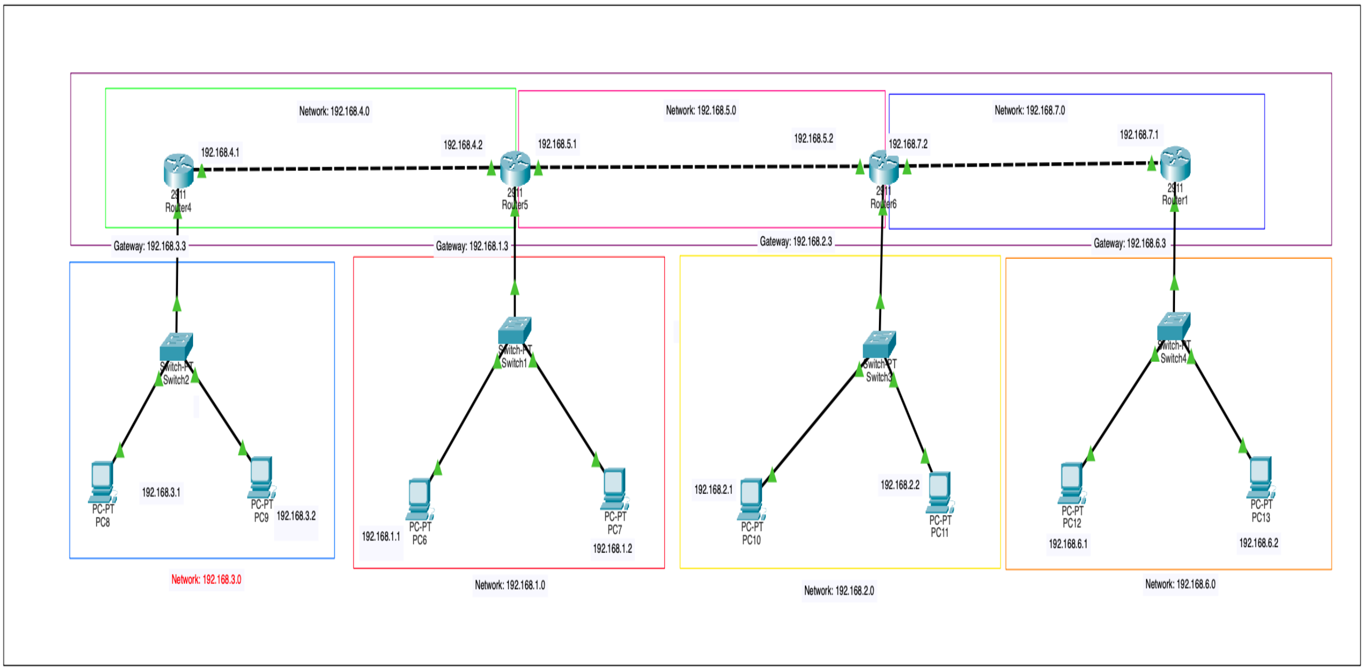
Question 1.

Develop Static Routing with 4 routers, 4 Switches & 8 Pc's. Make sure to highlight the Networks with their network ID's. IP Addresses should be written in text format as well.

Answer:



Static Routing Table:

Router	**Destination Network**	**Subnet Mask**	**Next Hop**
Router4	192.168.1.0	255.255.255.0	192.168.4.2
	192.168.2.0	255.255.255.0	192.168.4.2
	192.168.6.0	255.255.255.0	192.168.4.2
Router5	192.168.3.0	255.255.255.0	192.168.4.1
	192.168.2.0	255.255.255.0	192.168.5.2
	192.168.6.0	255.255.255.0	192.168.5.2
Router6	192.168.1.0	255.255.255.0	192.168.5.1
	192.168.3.0	255.255.255.0	192.168.5.1
	192.168.6.0	255.255.255.0	192.168.7.1
Router1	192.168.2.0	255.255.255.0	192.168.7.2
	192.168.1.0	255.255.255.0	192.168.7.2
	192.168.3.0	255.255.255.0	192.168.7.2

Network Table:

Network ID	Purpose	Devices Connected
192.168.1.0	LAN	Router5, PC6, PC7
192.168.2.0	LAN	Router6, PC10, PC11
192.168.3.0	LAN	Router4, PC8, PC9
192.168.4.0	WAN	Router4 to Router5
192.168.5.0	WAN	Router5 to Router6
192.168.6.0	LAN	Router1, PC12, PC13
192.168.7.0	WAN	Router6 to Router1

Question 2:

What is OSI model? What are the different layers of OSI model? Write the functionality, Data format and Protocols used at each layer.

Answer:

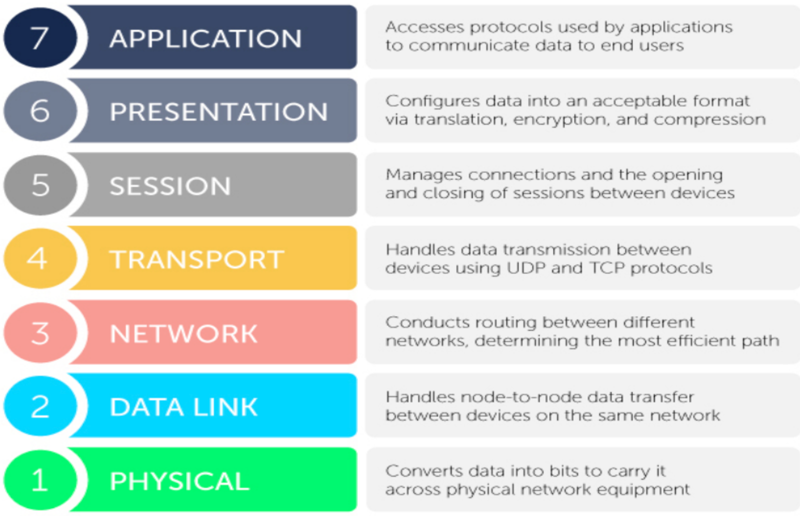
The OSI (Open Systems Interconnection) Model is a 7-layer conceptual reference model developed by the International Organization for Standardization (ISO) in 1984. It provides a standard framework that describes how data is transmitted from one computer to another over a network. The OSI model divides the communication process into seven layers, where each layer performs a specific function and communicates with the layer directly above and below it. This layered approach simplifies network design, troubleshooting, and protocol development while ensuring interoperability between devices manufactured by different vendors.

Objectives of the OSI Model

- Provides a standard for network communication.
- Enables interoperability between different hardware and software vendors.
- Simplifies network troubleshooting.
- Divides complex networking tasks into smaller manageable layers.
- Facilitates the development of new networking protocols.

Layers of the OSI Model

The OSI model consists of seven layers arranged from top (closest to the user) to bottom (closest to the hardware).



1. Physical Layer (Layer 1)

Functionality

- Responsible for transmitting raw binary data (0s and 1s) over the physical communication medium.
- Defines the electrical, mechanical and functional characteristics of the network.
- Specifies cables, connectors, voltage levels and transmission speed.
- Converts digital data into electrical, optical or radio signals.

PDU : Bits
Header: No header is added at this layer because it simply transmits bits over the transmission medium.
Protocols/Standards Used

- IEEE 802.3 (Ethernet Physical Standard)
- RS-232
- USB
- DSL
- ISDN
- Bluetooth
- Fiber Optic Standards

Devices

- Hub
- Repeater
- Cables
- Connectors

2. Data Link Layer (Layer 2)

Functionality

- Provides node-to-node communication between devices.
- Converts packets into frames.
- Performs framing.
- Uses MAC addresses for physical addressing.
- Detects transmission errors using CRC.
- Controls access to the communication medium.
- Performs flow control.

PDU: Frame
Header : The Data Link Layer adds:

- Source MAC Address
- Destination MAC Address

It also adds a Trailer containing the Frame Check Sequence (CRC) for error detection.

Protocols Used

- Ethernet (IEEE 802.3)
- PPP (Point-to-Point Protocol)
- HDLC
- Frame Relay
- ATM
- ARP

Devices

- Switch
- Bridge
- Network Interface Card (NIC)

<u>3. Network Layer (Layer 3)</u>

Functionality

- Provides logical addressing using IP addresses.
- Determines the best path between source and destination.
- Performs routing and packet forwarding.
- Handles fragmentation and reassembly of packets.
- Connects multiple networks together.

PDU: Packet

Header : The Network Layer adds the IP Header, which contains:

- Source IP Address
- Destination IP Address
- Time To Live (TTL)
- Protocol Number
- Fragment Information

Protocols Used

- IPv4
- IPv6
- ICMP
- RIP
- OSPF
- EIGRP
- BGP

Devices

- Router
- Layer-3 Switch

<u>4. Transport Layer (Layer 4)</u>

Functionality

- Provides end-to-end communication.
- Segments large data into smaller segments.
- Reassembles data at the destination.
- Performs error detection and recovery.
- Provides reliable communication using acknowledgements.
- Controls the flow of data.
- Uses port numbers to identify applications.
- Performs multiplexing and demultiplexing.

PDU: Segment (TCP) or Datagram (UDP)

Header: The Transport Layer adds a TCP/UDP Header, which contains:

- Source Port Number

- Destination Port Number
- Sequence Number
- Acknowledgement Number
- Window Size
- Checksum

Protocols Used

- TCP
- UDP
- SCTP

5. Session Layer (Layer 5)

Functionality

- Establishes communication sessions between applications.
- Maintains active sessions.
- Terminates completed sessions.
- Provides synchronization between sender and receiver.
- Supports checkpointing and recovery in case of failures.

PDU: Data

Header : The Session Layer adds a Session Header, which contains:

- Session ID
- Synchronization Information
- Checkpoint Information

Protocols Used

- NetBIOS
- RPC (Remote Procedure Call)
- PPTP
- SIP

6. Presentation Layer (Layer 6)

Functionality

- Translates data into a common format.
- Encrypts data before transmission.
- Decrypts received data.
- Compresses data to reduce transmission size.
- Decompresses received data.
- Converts character encoding formats.

PDU: Data

Header : The Presentation Layer adds a Presentation Header, which contains information related to:

- Data Format
- Encryption
- Compression
- Character Encoding

Protocols/Standards Used

- SSL/TLS
- JPEG
- PNG
- GIF
- MPEG
- ASCII
- Unicode

7. Application Layer (Layer 7)

Functionality
- Provides network services directly to end users.
- Supports web browsing.
- Supports email communication.
- Supports file transfer.
- Supports remote login.
- Provides access to network resources.

PDU: Data
Header : The Application Layer adds an Application Header, which contains application-specific information such as:
- HTTP Request Header
- FTP Commands
- SMTP Commands
- DNS Queries

Protocols Used
- HTTP
- HTTPS
- FTP
- SMTP
- POP3
- IMAP
- DNS
- DHCP
- SSH
- Telnet

Question 3:

What are the different Error Detection methods, Explain each with the help of proper examples.

Answer:

During data transmission, information may become corrupted due to noise, interference, attenuation, or hardware faults. As a result, one or more bits of the transmitted data may change, causing the receiver to receive incorrect information. To ensure the integrity of data, **error detection techniques** are used. These techniques add extra bits, called **redundant bits**, to the original data so that the receiver can determine whether an error has occurred during transmission.

The three most commonly used error detection methods are:

1. Parity Check
2. Checksum
3. Cyclic Redundancy Check (CRC)

1. Parity Check

Parity Check is the simplest error detection technique. In this method, an additional bit called the **parity bit** is added to the original data before transmission. This parity bit ensures that the total number of 1's in the transmitted data is either **even** or **odd**.

There are two types of parity:

- **Even Parity**
- **Odd Parity**

Example (Even Parity)

Suppose the sender wants to transmit the following 7-bit data:

Data: 1011001

Number of 1's in the data = 4. Since 4 is already an **even** number, the parity bit added will be **0**.

- Parity bit: 0
- Transmitted data: 10110010

Now suppose one bit changes during transmission.

Received data: 10110010

Now the number of 1's = 3, which is odd. Since even parity was expected, the receiver concludes that an error has occurred.

Example (Odd Parity)

Data: 1011001

Number of 1's = 4

To make the total number of 1's odd, the parity bit will be 1.

- Parity bit: 1
- Transmitted data: 10110011

Now the total number of 1's = 5 (Odd).

Advantages

- Simple to implement.
- Requires only one additional bit.
- Detects single-bit errors.

Disadvantages

- Cannot detect errors if an even number of bits change.
- Cannot correct errors.

2. Checksum

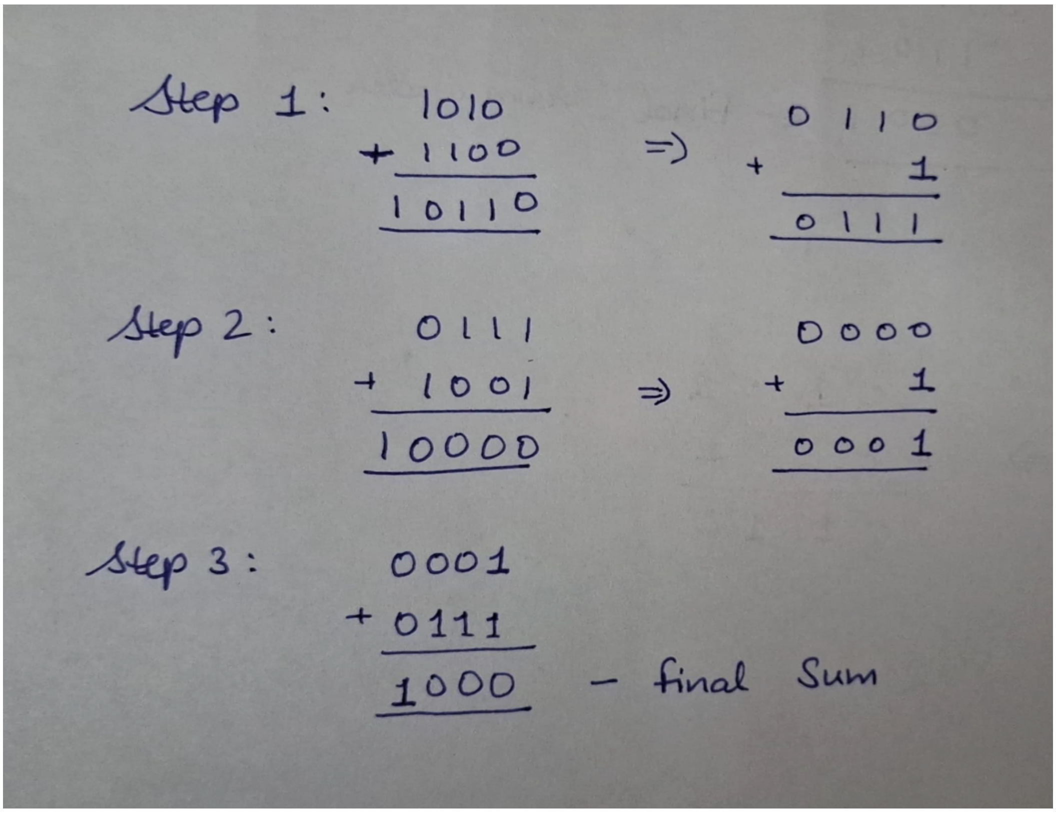
Checksum is another commonly used error detection technique. In this method, the sender divides the data into equal-sized blocks and adds all the blocks together using 1's complement arithmetic. The 1's complement of the final sum is called the checksum, which is transmitted along with the data.

At the receiver's end, the same addition is performed. If the final result is all 1's, the data is considered correct. Otherwise, an error is detected.

Example

Suppose the sender has the following four 4-bit data blocks: 1010, 1100, 1001 and 0111

Step 1: Add all the blocks



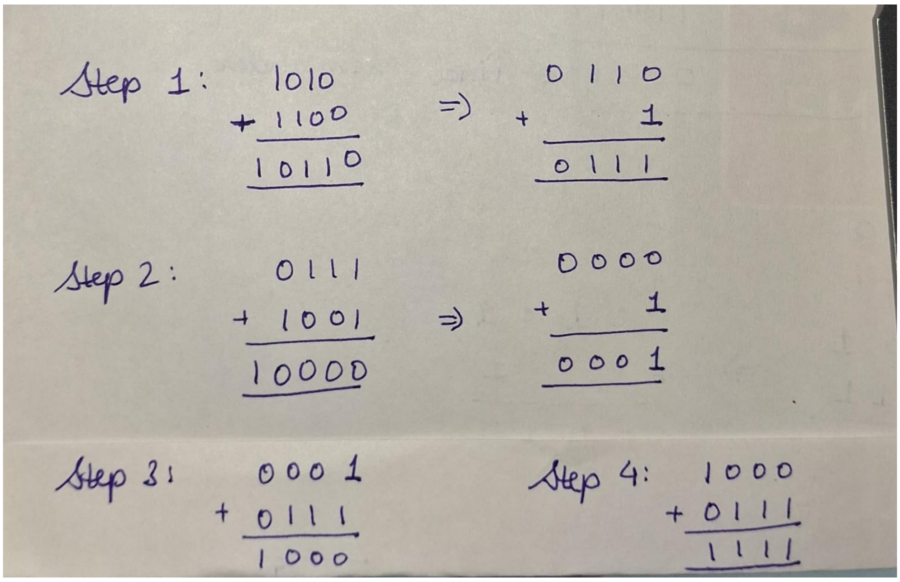
Step 2: Take the 1's Complement

Final Sum: 1000

1s Complement: 0111

The sender transmits: 1010, 1100, 1001, 0111, 0111 (checksum)

At the receiver's end, all the blocks including the checksum are added again.



- If the final result is 1111, the data is accepted.
- If the result is different, an error is detected.

Advantages

- More reliable than parity checking.
- Detects many common transmission errors.
- Widely used in TCP, UDP and IP protocols.

Disadvantages

- Cannot detect every possible error.
- Less reliable than CRC.

3. Cyclic Redundancy Check (CRC)

Cyclic Redundancy Check (CRC) is one of the most reliable error detection techniques. It treats the binary data as a polynomial and divides it by a predefined **generator polynomial** using **Modulo-2 division** (binary division without carry or borrow). The remainder obtained after the division is called the **CRC bits**, which are appended to the original data before transmission.

At the receiver's end, the same division is performed.

- If the remainder is 000, the data is accepted.
- If the remainder is non-zero, an error has occurred.

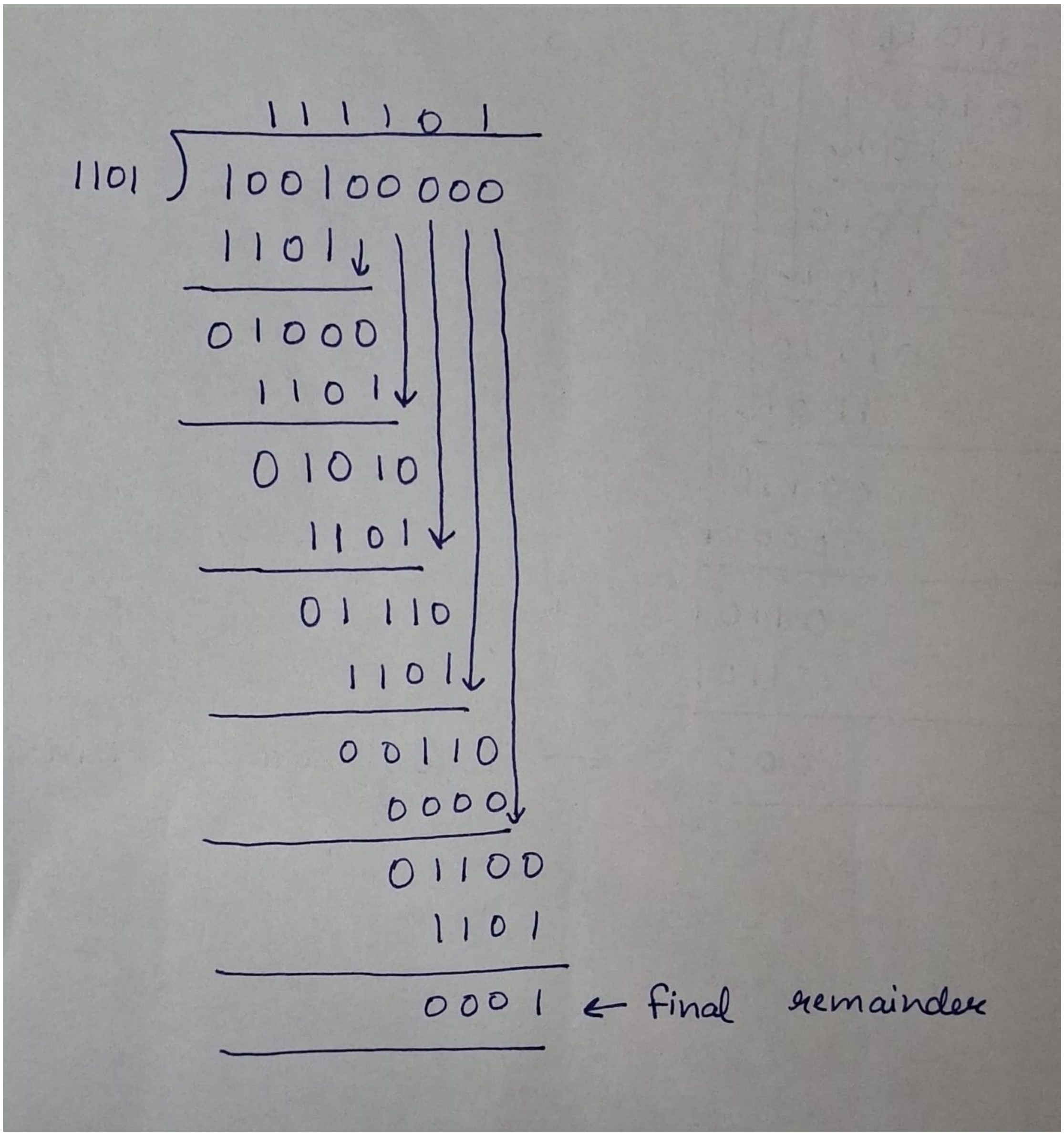
Example

Suppose Data: 100100 and Generator Polynomial is 1101

Since the generator contains 4 bits, append 3 zeros to the data.

Data: 100100000

Perform Modulo-2 Division.



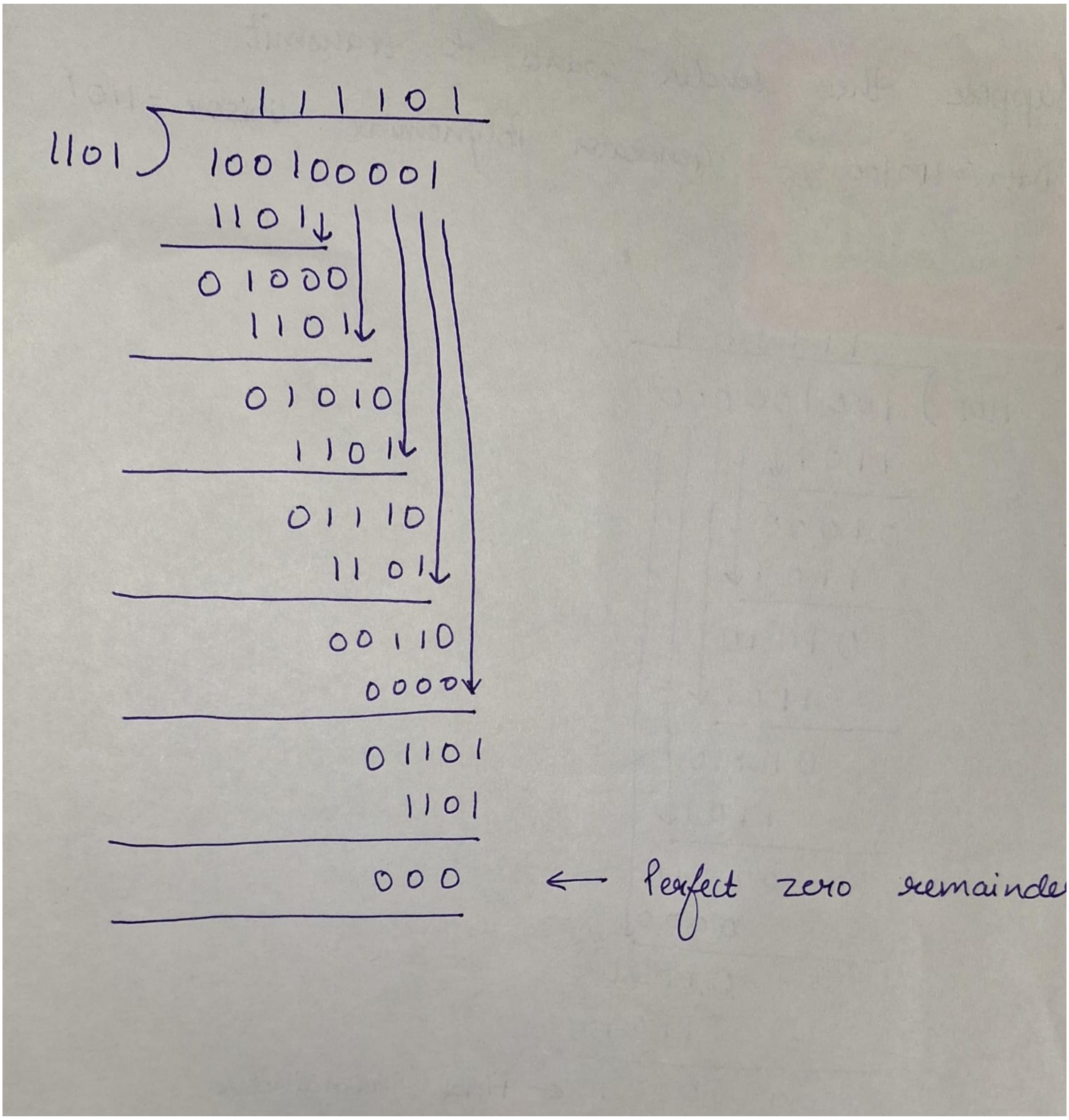
The remainder obtained after division is 001.

The sender appends this remainder to the original data. Therefore,

- Original Data: 100100
- CRC: 001
- Transmitted Data: 100100001

Receiver Side

The receiver again divides:



Since the remainder obtained is 000 so no error was detected. Otherwise, error would have been detected.

Advantages

- Highly reliable.
- Detects single-bit, multiple-bit and burst errors.
- Widely used in Ethernet, Wi-Fi, USB and storage devices.

Disadvantages

- More complex than parity check and checksum.
- Requires additional hardware or software support.